

Test Project: Student Report Card

Maximum marks	30 (+3 bonus)	Time allowed	60 minutes
Files to submit	index.html, style.css, script.js	Folder name	YourRollNo_YourName

Instructions

- This is an **individual** test. Build everything yourself, from an empty file.
- Use **only** what you have learned in class. No libraries, no frameworks, no copy-paste from the internet.
- Follow the names given in this paper **exactly** (ids, classes, function names). Spelling mistakes will cost you marks.
- Keep your three files in one folder. The CSS and the JavaScript must be **linked** to your HTML page.
- Test your page in the browser and check the **Console** (F12) before submitting.

The Problem

Your college wants a simple web page that shows the result of a student. The marks are fixed inside your JavaScript, and your program must **calculate** the total and the percentage and then **show them on the page** using the DOM. You will build the page (HTML), style it (CSS), and make it smart (JavaScript).

Part A: HTML, build the page structure (7 marks)

Create `index.html` with the following structure. Use these exact ids and classes.

Element	Id / Class	Content to write
title tag	-	Report Card
h1	id = school-title	Any college name of your choice
p	id = student-name	Name: (your own name)
p (two of them)	class = info	Branch: CSE and Year: 1st
h2	-	Marks
ul	id = marks	Contains three li items (next row)
li (three of them)	class = subject	Maths: 80 / Physics: 70 / Programming: 90
div	class = result-box	Contains the two p tags below
p (inside the div)	id = total	Total: --
p (inside the div)	id = percent	Percentage: --

The page must also be linked to `style.css` and `script.js`. Think carefully about **where** the script should be placed, so that the elements already exist when JavaScript looks for them.

Part B: CSS, style the page (7 marks)

In `style.css`, write rules that do all of the following. You choose the actual colours and sizes.

- Give the **body** a background colour and a font family.
- Style the h1 using its **tag** selector: centre the text and give it a colour.
- Style the student-name paragraph using its **id** selector: make it bold and larger.

- Style the `.info` paragraphs using their **class** selector: change the text colour.
- Style each `.subject` item: add a border and some padding.
- Style `.result-box`: add a border, padding, a background colour and a width. Give it some margin so it does not touch the other content.

Marks are given for using all three kinds of selectors correctly (tag, class and id) and for a neat result.

Part C: JavaScript, variables and functions (7 marks)

In `script.js`, write the following. Do not write any HTML numbers by hand into the page for the result: the page must get them from your functions.

- Store the three marks in **three variables** (80, 70 and 90) and the maximum total (300) in another variable. Decide carefully which keyword is right for each. Also create a variable named `total` that starts at 0 and will change later.
- Write a **normal function** `calculateTotal(m1, m2, m3)` that **returns** the sum of the three marks.
- Write an **arrow function** `calculatePercentage(total, maxMarks)` that **returns** the percentage.
- Call your functions with your variables and store the results.

Part D: DOM, connect JavaScript to the page (7 marks)

Using the DOM, complete all of these in `script.js`:

#	What to do	Method to use
1	Select the <code>total</code> paragraph and change its text to <i>Total: 240</i> (use your calculated value, do not type 240).	<code>getElementById</code>
2	Select the <code>percent</code> paragraph and change its text to <i>Percentage: 80%</i> (use your calculated value).	<code>getElementById</code>
3	Select all <code>subject</code> items. Print how many there are, and print the text of the third one.	<code>getElementsByClassName</code>
4	Select all <code>p</code> tags. Print how many there are. (It should be 5 if your HTML follows Part A.)	<code>getElementsByTagName</code>
5	Select the first <code>.info</code> paragraph and print its text.	<code>querySelector</code>
6	Select the <code>#percent</code> paragraph and change its text colour to green.	<code>querySelector</code>

Part E: Understanding (2 marks)

At the end of `script.js`, write two short **comments** (lines starting with `//`) in your own words:

1. What is the **document** in the DOM?
2. Why does `getElementsByClassName` need a position number (like `[2]`) to get one item, while `getElementById` does not?

Expected Result

The page should show **Total: 240** and **Percentage: 80%** (in green). Your console should print something like this:

```
Subjects found: 3
Third subject: Programming: 90
Paragraphs found: 5
First info: Branch: CSE
```

Your wording and your branch may be different. The numbers and structure must be correct.

Bonus (+3 marks)

The teacher gives **5 grace marks** in Physics. Write an arrow function `addGrace(marks, grace)` that returns the new marks, then use it to recalculate and update the total and the percentage on the page.

Marking Scheme

Part	What is checked	Marks
A: HTML	Correct structure, exact ids and classes, files linked, script placed correctly	7
B: CSS	All rules present, tag, class and id selectors used correctly, neat look	7
C: Variables and functions	Right keyword for each variable, one normal and one arrow function, both return correctly	7
D: DOM	All six DOM tasks work with the correct method	7
E: Understanding	Two comments explained in own words	2
Total		30

Tip: Build one part at a time. Open the page and the console after each part, so a small mistake does not turn into a big one.